

Chenyu Fang, Ph.D.

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SUMMARY

Machine Learning Engineer (Ph.D.) with 3+ years of applied ML/research experience taking models from data to deployment: time-series forecasting, rare-event classification, and LLM evaluation and fine-tuning. Strongest in experiment design, honest baselines, and model serving.

TECHNICAL SKILLS

ML and languages: Python, PyTorch, scikit-learn, XGBoost, LightGBM, GNNs, transformers (TFT, ParT), SQL, C++
LLM systems: LoRA fine-tuning (Hugging Face, PEFT), distillation, LLM evaluation, vLLM, RAG, LangGraph, MCP
Deployment and data: FastAPI, Docker, Kubernetes (GKE, kind), ONNX, Azure (Container Apps), AWS, GCP (Vertex AI, Cloud Run), PostgreSQL/pgvector, GitHub Actions, pytest, Linux, Git

SELECTED PROJECTS

Gridcast | NYISO Day-Ahead Load Forecasting

Demo: gridcast.cyfang.org | Code: github.com/chenyufang-data/gridcast

- Trained a global Temporal Fusion Transformer across 11 NYISO zones and the statewide total; served it as ONNX through FastAPI on GCP with daily ingestion, self-scoring against NYISO, and a LightGBM fallback.
- Cut average hourly error (MAPE) 53% below the best naive baseline in a 12-month backtest, on par with NYISO's own day-ahead forecast (**4.80% vs. 4.85%**; difference not significant under a paired bootstrap).
- Ran 20+ experiments; conformal calibration raised LightGBM's hours inside the 80% forecast range from 57% to 77%.
- Deployed it on Kubernetes (kind in CI on every push, a 72-hour GKE Autopilot trial) with a digest-pinned model image.

MLE/AIE Interview Coach

Demo: coach.cyfang.org | Code: github.com/chenyufang-data/mle-aie-interview-coach

- Trained scikit-learn and LoRA-tuned Qwen3-4B graders on 598 Claude-labeled synthetic answers (~20% held out per seed); Qwen3-4B agreed with Claude's grades at **0.94 on a 0-1 scale (QWK)**, beating scikit-learn on every seed.
- Benchmarked vLLM serving at 30 ms per grading call (p95); the deployed app enforces rate limits and usage budgets.
- Raised how often the right passage ranks in the top 5 (**Recall@5, 72% to 84%**) with BM25 + BGE search, and chose DeepSeek for hosted grading at ~1/200th Claude's cost with 94% of scores within one point.

AI Tender Evaluation Assistant

02/2026 - Present

Team project, Techlent Machine Learning Engineer Fellowship (training program)

Demo: tender-eval-ai.github.io/tender-evaluation-assistant | Code: github.com/tender-eval-ai/tender-evaluation-assistant

- Co-built an LLM bid-review system with a second engineer; I owned the FastAPI backend, LLM gateway and bid checks. Models only read, a deterministic rule engine decides, and every value is cited to its page.
- Built Postgres-backed jobs with checkpoints, approval pauses and rate-limit backoff; chose Procrastinate over LangGraph after a 12-scenario fault-injection harness showed about 5 s recovery and **0 lost jobs**.
- Verified every extracted value by text-layer match or an independent second read, sending failures to human review; on a synthetic tender, rule drafting matched 15/15 checklist items.

EXPERIENCE

University of Oklahoma

Norman, OK

Research Assistant (part-time)

02/2026 - Present

Rare-Signal Benchmark: XGBoost vs. LorentzNet vs. Particle Transformer

Code: github.com/chenyufang-data/bsm-higgs-ml-benchmark

- Raised the effective sample size of the simulated data 9-19x by weighting events with tagging probabilities instead of random accept/reject, keeping expected yields unbiased.
- Benchmarked XGBoost, a re-implemented LorentzNet and a Particle Transformer over three seeds under rules fixed in advance; ML beat the rule-based baseline by **9-21% in expected significance**, and both networks beat XGBoost's AUC.
- LorentzNet matched the transformer at a tenth of its size; XGBoost trailed by under 0.01 AUC at ~1/100th the cost.

PhD Researcher

08/2018 - 12/2025

Code: github.com/chenyufang-data/hep-ml-pipeline

- Built Python/C++ pipelines and XGBoost classifiers for multi-million-event simulated datasets; used SHAP and feature ablations to evaluate signal separation, recovering **41.2% of signal at a 7.7% false-positive rate**.

Research outputs: Two first-author physics preprints (2025): [arXiv:2511.08526](https://arxiv.org/abs/2511.08526), [arXiv:2511.19604](https://arxiv.org/abs/2511.19604).

EDUCATION

University of Oklahoma - PhD in Physics

Norman, OK | 08/2018 - 12/2025

Huazhong University of Science and Technology - B.S. in Physics

Wuhan, China | 09/2014 - 06/2018